

BHAVISHAYA BANSAL

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SUMMARY

Software Engineer specializing in scalable backend systems with 2 years of experience building **high-throughput, concurrency-safe systems** and **low-latency APIs** in production environments. Experienced in designing *scalable resource management platforms*, implementing **idempotent services**, and ensuring **data consistency under high concurrency**. Proven ability to *optimize system performance and reliability*, with hands-on experience integrating **LLM-based solutions**, **real-time processing pipelines**, and **distributed backend architectures**. Strong in **Java**, **Node.js**, **REST APIs**, and **relational databases**.

EXPERIENCE

Software Engineer – Backend

Nov 2024 – Present

Voxomos Systems - Product-Based Startup

Noida, U.P.

Resource Management Server

- Designed and built a **Java-based Resource Management System** to programmatically allocate and release SIP lines across *distributed telephony campaigns*, improving resource utilization by ~60%.
- Implemented **low-latency**, idempotent **REST APIs** backed by an immutable **MySQL** debit-credit ledger, ensuring consistency under high concurrency and handling race conditions at scale.
- Engineered thread-safe allocation logic using `BlockingQueue` and priority queues based on CPS, urgency, and availability, reducing queue latency by ~35% and call failures by ~80%.

AI-Powered Email Phishing Detection System

- Orchestrated a **real-time, multi-layer email threat detection system** to analyze URLs, content, and attachments using LLM-based classification and secure sandboxing.
- Integrated **Llama 3-based** inference with structured outputs and implemented **Docker-based** sandbox pipelines (URL expansion, CDR), improving detection accuracy to ~85-90%.
- Architected modular backend services and automated decision engine, enabling scalable processing and handling ~40-50 emails/min with parallel execution.

Multilingual Communication System (LLM-Driven)

- Optimized and scaled backend-driven communication flows, by integrating *LLM-based multilingual logic*, powering high-volume outbound campaigns across **8+ languages**.
- Created **Python** automation scripts to streamline repetitive workflows and performed early-stage issue detection and debugging on **Linux** systems, ensuring stability during large-scale campaign rollouts.

Software Developer

Feb 2024 – Oct 2024

ZenTrades Inc. - SaaS Product-Based Startup

Pune, Maharashtra

Excel Report Format Correction & Standardization

- Designed and optimized **Node.js REST APIs** for report workflows, adding caching to reduce response time by ~40%.
- Refactored backend data grouping and aggregation logic to enable configurable **multi-dimensional reporting** at scale.
- Scaled **React.js** components for nested table rendering and interactive report views, improving report usability, reducing time-to-insight by ~25% and enabling flexible exploration of large datasets.

Proposal Copilot

- Built scalable backend **Express.js** APIs to generate proposals based on user's inputs, reducing manual effort by ~80%.
- Defined API contracts and backend data flows in collaboration with product and frontend teams.
- Led frontend development of an AI-driven proposal generation system using **React.js**, translating backend APIs and product requirements into an intuitive UI that accelerated proposal creation workflows.

TECHNICAL SKILLS

Languages: Java, JavaScript, SQL, Python, TypeScript

Backend & Systems: Node.js, REST API Design, Microservices, Express.js, MySQL, MongoDB, Asterisk, Linux

C.S. Fundamentals: DSA, Concurrency & Multithreading, Distributed Systems, Database Internals, System Design

AI & LLM Systems: LLM Integration (Llama 3), NLP Classification, Prompt Engineering, Real-time Pipelines

Security & Processing: Secure Sandboxing, CDR, URL Analysis, Threat Detection

Tools, Infra & AI Agents: Docker, Git, Bash, Postman, VS Code, Asterisk CLI, Claude Code, GPT Codex

EDUCATION

Vellore Institute of Technology

Sep 2020 – Jul 2024

B.Tech - Computer Science and Engineering - 8.54/10.0 GPA

India